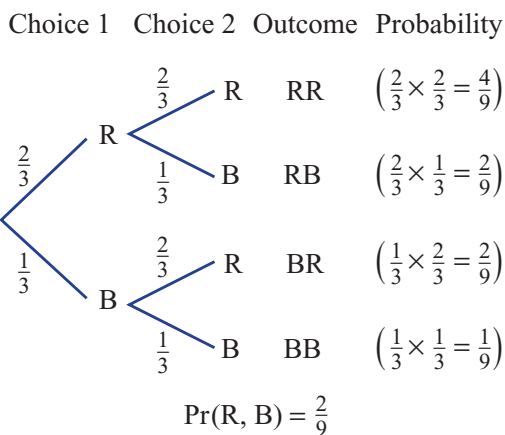
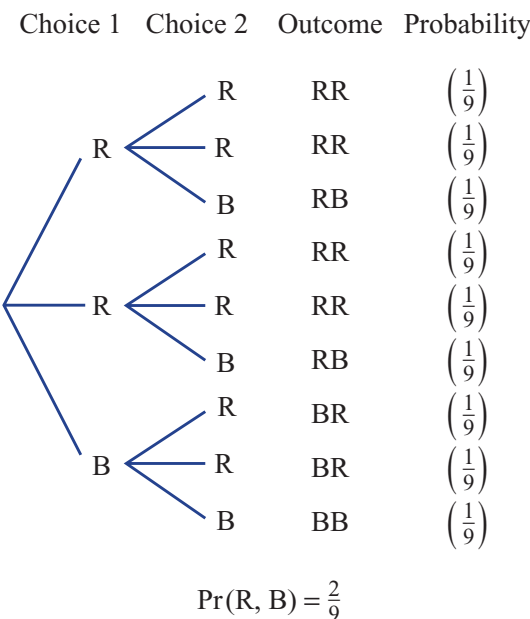


4.6 Using tree diagrams



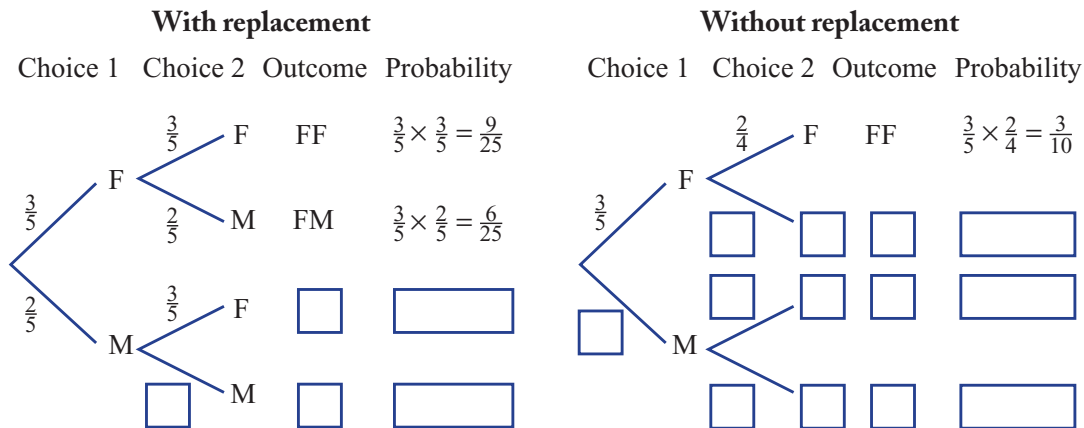
Tree diagrams can also be used to help list outcomes for multiple events. Suppose that a bag contains two red counters and one blue counter and that two counters are selected at random with replacement. One way to display the outcomes is with a tree diagram in which all equally likely outcomes are listed in columns, as shown below left. A more efficient way, however, is to group similar outcomes and write their corresponding probabilities on the branches, as shown below right.



In the tree diagram on the right, the probability of each outcome is obtained by multiplying the branch probabilities. This also applies when selection is made without replacement.

► Let's start: Trees with and without replacement

Suppose that two selections are made from a group of 2 male and 3 female workers to complete two extra tasks.

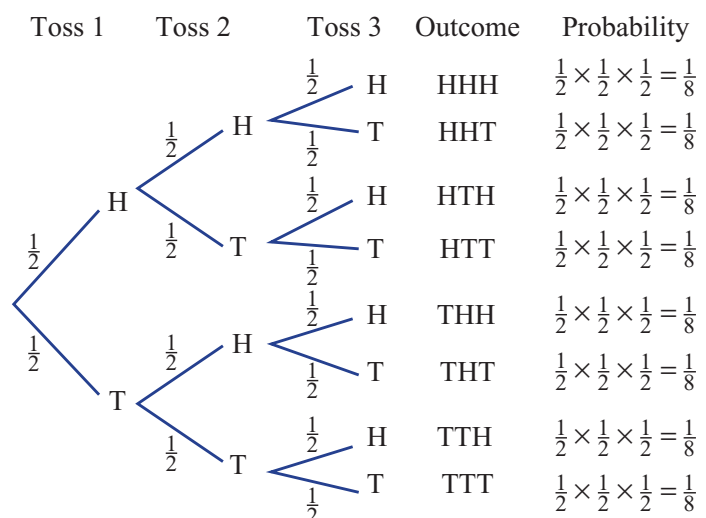


- Complete these two tree diagrams to show how these selections can be made, both with and without replacement.
- Explain where the branch probabilities come from on each branch of the tree diagrams.
- What is the total of all the probabilities on each tree diagram?

- Tree diagrams can be used to list the sample space for experiments involving two or more components.
 - Branch probabilities are used to describe the chance of each outcome at each step.
 - The probability of each outcome for the experiment is obtained by multiplying the branch probabilities.
 - Branch probabilities will depend on whether selection is made with or without replacement.
 - For *with replacement*, probabilities do not change.
 - For *without replacement*, probabilities do change.

Exercise 4F

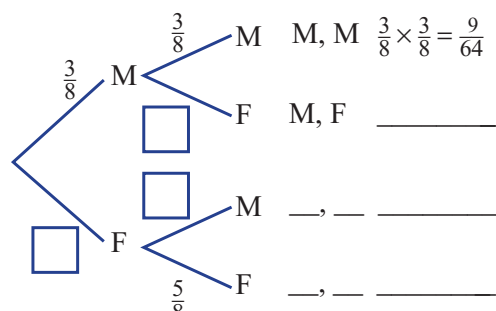
- 1 A coin is tossed three times and a head or tail is obtained each time.
- a How many outcomes are there?
 - b What is the probability of the outcome HHH?
 - c How many outcomes obtain:
 - i 2 tails?
 - ii 2 or 3 heads?
 - d What is the probability of obtaining at least one tail?



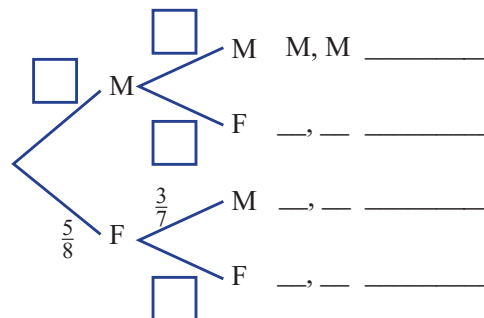
4F

- 2 Two prizes are awarded to a group of 3 male (M) and 5 female (F) candidates. Copy and complete each tree diagram. Include the missing branch probabilities and outcome probabilities.

a with replacement



b without replacement



For 'without replacement' the second selection is out of 7, not 8.



- 3 A box contains 2 white (W) and 3 black (B) counters.

a A single counter is drawn at random. Find the probability that it is:

i white

ii black

b Two counters are now drawn at random. The first one is replaced before the second one is drawn. Find the probability that the second counter is:

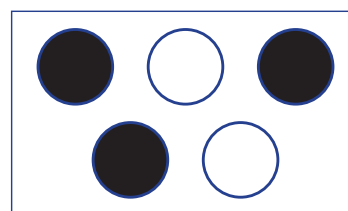
i white

ii black

c Two counters are drawn and the first counter is not replaced before the second one is drawn. If the first counter is white, find the probability that the second counter is:

i white

ii black



After one white counter is taken out, how many of each remain?



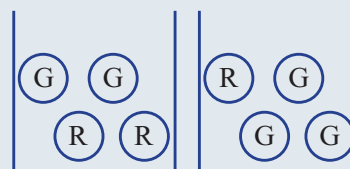
Fluency

Example 10 Constructing a tree diagram for multiple events

Boxes A and B contain 4 counters each. Box A contains 2 red and 2 green counters and box B contains 1 red and 3 green counters. A box is chosen at random and then a single counter is selected.

- What is the probability of selecting a red counter from box A?
- What is the probability of selecting a red counter from box B?
- Represent the options available as a tree diagram that shows all possible outcomes and related probabilities.
- What is the probability of selecting box B and a red counter?
- What is the probability of selecting a red counter?

Box A Box B



Solution

a $\Pr(\text{red from Box A}) = \frac{2}{4} = \frac{1}{2}$

b $\Pr(\text{red from Box B}) = \frac{1}{4}$

Explanation

Two of the 4 counters in box A are red.

One of the 4 counters in box B is red.

c

	Box	Counter	Outcome	Probability
$\frac{1}{2}$	A	$\frac{1}{2}$ red	(A, red)	$\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$
		$\frac{1}{2}$ green	(A, green)	$\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$
$\frac{1}{2}$	B	$\frac{1}{4}$ red	(B, red)	$\frac{1}{2} \times \frac{1}{4} = \frac{1}{8}$
		$\frac{3}{4}$ green	(B, green)	$\frac{1}{2} \times \frac{3}{4} = \frac{3}{8}$

First selection is a box followed by a counter.

Multiply each of the probabilities along the branch pathways to find the probability of each outcome.

d $\Pr(\text{B, red}) = \frac{1}{2} \times \frac{1}{4}$
 $= \frac{1}{8}$

The probability of choosing box B is $\frac{1}{2}$ and a red counter from box B is $\frac{1}{4}$, so multiply the probabilities for these two outcomes together.

e $\Pr(1 \text{ red}) = \frac{1}{2} \times \frac{1}{2} + \frac{1}{2} \times \frac{1}{4}$
 $= \frac{1}{4} + \frac{1}{8}$
 $= \frac{3}{8}$

The outcomes (A, red) and (B, red) both contain 1 red counter, so add the probabilities for these two outcomes together.

- 4** Boxes A and B contain 4 counters each. Box A contains 1 yellow and 3 orange counters and box B contains 3 yellow and 1 orange counter. A box is chosen at random and then a single counter is selected.

- a** If box A is chosen, what is the probability of selecting a yellow counter?
b If box B is chosen, what is the probability of selecting a yellow counter?
c Represent the options available by completing this tree diagram.

	Box	Counter	Outcome	Probability
$\frac{1}{2}$	A	$\frac{1}{4}$ Yellow	(A, Yellow)	$\frac{1}{2} \times \frac{1}{4} = \frac{1}{8}$
		☐ Orange	_____	_____
$\frac{1}{2}$	B	☐ Yellow	_____	_____
		☐ _____	_____	_____

- d** What is the probability of selecting box B and a yellow counter?
e What is the probability of selecting 1 yellow counter?

For part **e**, add the probabilities for both outcomes that have a yellow counter.



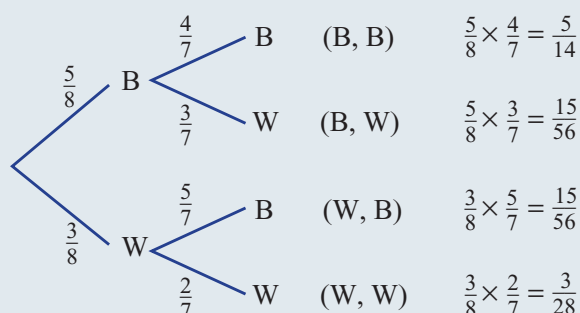
Example 11 Using a tree diagram without replacement

A bag contains 5 blue (B) and 3 white (W) marbles and two marbles are selected without replacement.

- a** Draw a tree diagram showing all outcomes and probabilities.
b Find the probability of selecting:
i a blue marble followed by a white marble (B, W)
ii 2 blue marbles
iii exactly one blue marble
c If the experiment was repeated with replacement, find the answers to each question in part **b**.

Solution**Explanation**

a Selection 1 Selection 2 Outcome Probability



After one blue marble is selected there are 7 marbles remaining: 4 blue and 3 white.

After one white marble is selected there are 7 marbles remaining: 5 blue and 2 white.

$$\begin{aligned} \text{b i } \Pr(B, W) &= \frac{5}{8} \times \frac{3}{7} \\ &= \frac{15}{56} \end{aligned}$$

Multiply the probabilities on the (B, W) pathway.

$$\begin{aligned} \text{ii } \Pr(B, B) &= \frac{5}{8} \times \frac{4}{7} \\ &= \frac{5}{14} \end{aligned}$$

Only 4 blue marbles remain after the first selection. Multiply the probabilities on the (B, B) pathway.

$$\begin{aligned} \text{iii } \Pr(1 \text{ blue}) &= \frac{5}{8} \times \frac{3}{7} + \frac{3}{8} \times \frac{5}{7} \\ &= \frac{15}{28} \end{aligned}$$

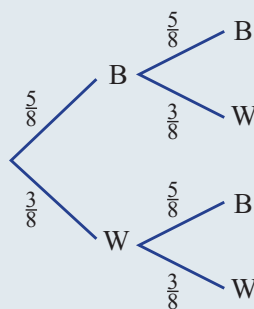
The outcomes (B, W) and (W, B) both have one blue marble. Multiply probabilities to find individual probabilities, then sum for the final result.

$$\begin{aligned} \text{c i } \Pr(B, W) &= \frac{5}{8} \times \frac{3}{8} \\ &= \frac{15}{64} \end{aligned}$$

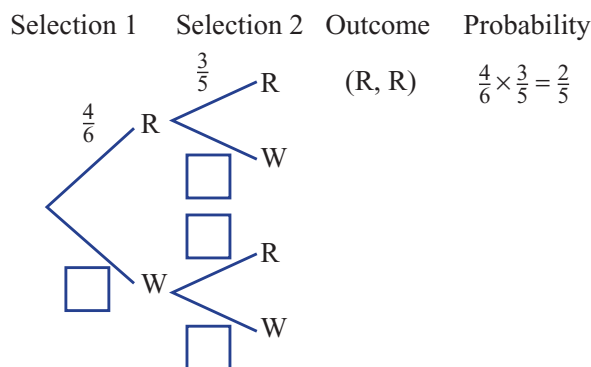
When selecting objects with replacement, remember that the number of marbles in the bag remains the same for each selection.

$$\begin{aligned} \text{ii } \Pr(B, B) &= \frac{5}{8} \times \frac{5}{8} \\ &= \frac{25}{64} \end{aligned}$$

$$\begin{aligned} \text{iii } \Pr(1 \text{ blue}) &= \frac{5}{8} \times \frac{3}{8} + \frac{3}{8} \times \frac{5}{8} \\ &= \frac{15}{32} \end{aligned}$$

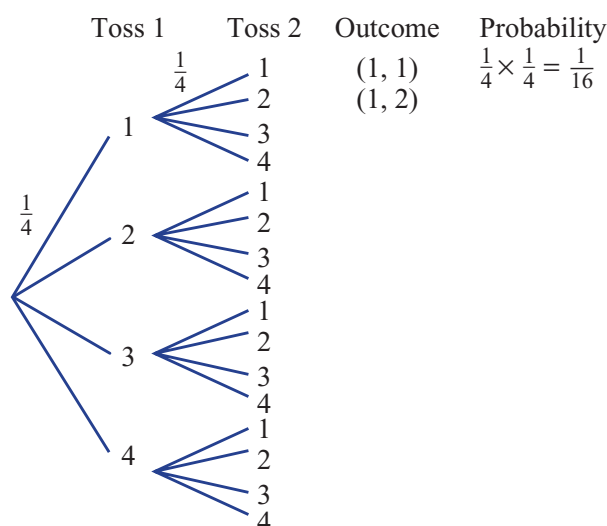


- replacement, find the answers to each question in part **a**.



-
- A probability tree diagram for a two-stage experiment. The first stage has two outcomes: M (top branch) and F (bottom branch), with a probability of $\frac{4}{7}$ for F. The second stage branches from each first-stage outcome. From M, there are two outcomes: M (top) and F (bottom). From F, there are two outcomes: M (top) and F (bottom), with a probability of $\frac{3}{6}$ for F.

- iii less than or equal to 5

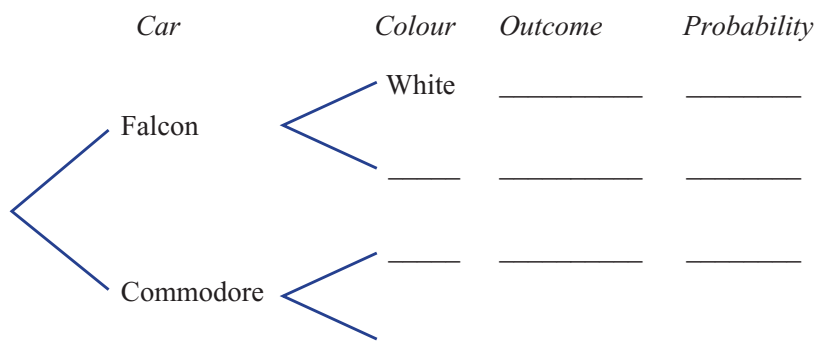


Problem-solving and Reasoning

4F

- 8 As part of a salary package a person can select either a Falcon or a Commodore. There are 3 white Falcons and 1 silver Falcon and 2 white Commodores and 1 red Commodore to choose from.

a Complete a tree diagram showing a random selection of a car type, then a colour.



b Find the probability that the person chooses:

- i a white Falcon ii a red Commodore iii a white car
iv a car that is not white v a silver car or a white car vi a car that is not a Falcon nor red

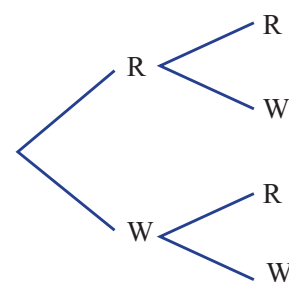
- 9 Two bottles of wine are randomly selected for tasting from a box containing 2 red and 2 white wines. Use a tree diagram to help answer the following.

a If the first bottle is replaced before the second is selected, find:

- i $\text{Pr}(2 \text{ red})$ ii $\text{Pr}(1 \text{ red})$
iii $\text{Pr}(\text{not two white})$ iv $\text{Pr}(\text{at least one white})$

b If the first bottle is not replaced before the second is selected, find:

- i $\text{Pr}(2 \text{ red})$ ii $\text{Pr}(1 \text{ red})$
iii $\text{Pr}(\text{not two white})$ iv $\text{Pr}(\text{at least one white})$



Rainy days



- 10 Imagine that the probability of rain next Monday is 0.2. The probability of rain on a day after a rainy day is 0.85, while the probability of rain on a day after a non-rainy day is 0.1.

a Next Monday and Tuesday, find the probability of having:

- i two rainy days ii exactly one rainy day
iii at least one dry day

b Next Monday, Tuesday and Wednesday, find the probability of having:

- i three rainy days ii exactly one dry day
iii at most two rainy days

